Sahaj Bhadja

(213) 284-4651 bhadja@usc.edu sahaj-bhadja sahajbhadja sahaj-bhadja

Summary

Full-stack engineer with 2+ years shipping **AI automation**, **Python**, **TypeScript/React**, analytics, and customer-facing workflows. Combines production ownership, technical documentation, stakeholder communication, and disciplined review of AI-generated outputs.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Web Technologies, Database Systems, Advanced NLP, Analysis of Algorithms

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Object-Oriented Programming, Data Retrieval, Operating Systems

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*

- Architected a multi-tenant **AI platform** using Azure SQL and Blob Storage for 500+ users, reducing dashboard latency from 1.2 seconds to 350 milliseconds through query and indexing optimization.
- Built ETL-style integrations across FastAPI, Supabase, SQL Server, and cloud storage, processing 5,000+ monthly jobs with reconciliation, error handling, documentation, and CI/CD-backed releases achieving 99%+ production reliability.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*

- Engineered a production AI grading platform using GPT-4o, Claude, Qwen-VL, and OCR to evaluate 10,000+ daily submissions with 90%+ consistency while serving clear feedback to 150+ instructors.
- Designed secure PDF, DOCX, and ZIP ingestion with concurrent Python processing, model-usage logging, REST APIs, AWS RDS, and CI/CD, reducing instructor feedback turnaround 60% under production workloads.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*

- Built a Python evaluation pipeline processing 100,000+ records, improving model quality 18% through reproducible experiments, automated metrics, controlled configurations, performance tracing, and structured analysis of failure modes.
- Collaborated on model evaluation across complex scenarios, improving stability 15% while inspecting outputs, debugging failure patterns, validating robustness, and communicating tradeoffs between learned behavior and expected user outcomes.

Aasma Technology Solutions **India**
Software Developer Engineer – SecureSync *August 2023 – July 2024*

- Engineered cross-platform data-sharing software with automated builds and 20+ unit and integration test suites using Jenkins, PyTest, and Mockito while collaborating with Microsoft and GitHub stakeholders in Agile sprints.
- Optimized large-scale API integrations using OOP, AWS Glue, and S3, improving response times 30%; led ETL workflow improvements that increased pipeline efficiency 68% and reduced technical debt 20%.

Aasma Technology Solutions **India**
Full Stack Developer Intern – VibrantChem *May 2023 – August 2023*

- Developed a MERN-based **CRM** with WebSockets, REST APIs, and eight microservices supporting customer, lead, billing, permissions, and notification workflows while reliably synchronizing more than two million records.
- Integrated Google Analytics and automated Jenkins deployments, reducing resource waste 42%, increasing lead conversions 15%, and cutting deployment time 50% while consistently maintaining 95%+ successful build validation.

Skills

Languages: Python, TypeScript, JavaScript, SQL, Java, C++, Bash, HTML, CSS
Frontend & Design: React.js, Next.js, Tailwind CSS, Material UI, Figma, responsive design, dashboards, data visualization
Backend & Data: FastAPI, Flask, Node.js, REST APIs, WebSockets, PostgreSQL, MySQL, SQL Server, MongoDB, Firebase, ETL
AI & Automation: Claude, Codex, Cursor, LLM APIs, prompt workflows, OCR, RAG, semantic search, source-grounded generation, human review
Cloud & Delivery: AWS, Azure, GCP, Vercel, Render, Docker, Kubernetes, Jenkins, GitHub Actions, Git, CI/CD, PyTest
Practices: Technical documentation, analytics, requirements gathering, stakeholder communication, Agile, testing, code review, production support, monitoring

Projects

GTM Signal Hub | *Next.js, TypeScript, Python, FastAPI, SQLite, PyTest*

- Built a deterministic GTM analytics and RFP workflow that normalizes synthetic CRM exports, exposes reviewed metrics, rejects unsupported answers, cites approved sources, and passes three automated grounding tests.

HMIS | Live | *TypeScript, Healthcare APIs, Sentry, Twilio*

- Built a healthcare workflow platform with consent-based records access, scheduling, notifications, dashboards, and Sentry monitoring, reducing administrative workload 30% while enabling doctors to retrieve patient histories twice as fast.

PiXMod | Live | *Next.js, AI Models, MongoDB, Stripe*

- Built a Next.js AI SaaS platform combining semantic image search, segmentation, generative editing, MongoDB metadata, and Stripe subscriptions, processing 1,000+ edited assets while increasing premium conversion 18%.